

Nikhilan Velumani

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EDUCATION

Master of Science, Information Technology & Management, *The University of Texas at Dallas* 05/2025

Relevant coursework: Tech and New Product Development, Data Visualization, Operations Management, Business Data Warehousing, System Analysis and Product Management

Bachelor of Technology, Computer Science & Engineering, *SRM IST, India* 05/2020

PROFESSIONAL EXPERIENCE

TransUnion, Chennai, India 11/2020 – 07/2023
Associate Developer

- Championed the roadmap for an in-house address standardization framework, collaborating with teams to align with business objectives, saving \$120,000 annually by eliminating third-party software.
- Designed a Python-based archival framework, boosting data efficiency by 25% and cutting storage costs by 30%, contributing to a 20% operational expense reduction OKR.
- Streamlined data processes in collaboration with cross-functional teams, reducing manual effort by 10 hours daily.
- Implemented a comprehensive data archiving and purging framework using Ab Initio, decreasing database maintenance time by 40%.
- Acted as a liaison with business partners to evaluate the impact of new requests on existing systems and processes.
- Conducted reverse engineering on existing graphs and stored procedures and developed generic Ab Initio ETL graphs to address business requests.
- Led the end-to-end testing of ETL applications, including system and integration testing.

TransUnion, Chennai, India 02/2020 – 05/2020
Analyst (Intern)

- Contributed to creating and standardizing Ab Initio ETL graphs for the CoP (Community of Practice) team, enhancing generalization and compliance within the company.
- Assisted job scheduling within the Control Centre (ETL) and facilitated object deployments using Jenkins.

ACADEMIC PROJECT EXPERIENCE

Medical Diagnosis using Digital Data Twin, *UT Dallas* 07/2024

- Engineered a Medical Diagnosis Digital Data Twin, creating a virtual patient replica for real-time health monitoring and predictive diagnostics using machine learning models and AWS cloud services.
- Developed an interactive dashboard prototype simulating drug effects on patient digital twins, integrating EHRs, wearables, and medical imaging data to enable personalized treatment plans.

Mobile Surveillance Agency Data Infrastructure Transformation, *UT Dallas* 05/2024

- Migration from a legacy on-premises system to an AWS cloud-based platform, enhancing real-time surveillance and predictive analytics using AWS S3, Glue, Kinesis Data Firehose, and SageMaker.
- Ensured data security and compliance with AWS IAM, Guard Duty, and CloudTrail, resulting in a 25% reduction in data processing time and significant cost savings.

Weather Data Pipeline, *UT Dallas* 01/2024

- Engineered an optimized AWS cloud architecture for weather data, implementing Glue crawlers for automated ETL and fine-grained IAM policies, resulting in 20% reduced processing time and 80% faster SQL performance on 2TB datasets queried through Athena.

Network Analysis of Facebook Pages, *UT Dallas* 12/2023

- Conducted network analysis on a large Facebook Page dataset of 100k users in Gephi, revealing key nodes and clusters using centrality and modularity metrics.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, R, SQL, Unix/Unix Scripting, SparkSQL, HiveQL

Visualization: Tableau, Microsoft Power BI, D3, Altair, Gephi, SAP BI

Data: Ab Initio Suite (GDE/Express IT/CC), Autosys, Dremio, Splunk, Snowflake, Hadoop, Hive

Tools: Visio, Rally, MS Office Suite, Qualtrics, AWS, DataBricks, Microsoft Power Automate, UiPath

Skills: System Analysis and Project Management, Data Analytics, Agile, Operations Management

Certifications: AWS Certified Developer – Associate, AWS Certified Cloud Practitioner

LEADERSHIP & ORGANIZATION EXPERIENCE

Student Government (Graduate and International Affairs Chair), *UT Dallas* 09/2023 – Present

Orientation Leader, *UT Dallas* 03/2024 – Present

Co-leader (CSR & Employee Engagement), *TransUnion* 12/2020 – 06/2023